Note that this schedule can change, so check it frequently!

Last update: September 23, 2026

Date		Time	Room	Lecturer	Topic	Sections of the textbook	Notes and additional readings
Tuesday	15/9	8:15 - 10:15	6.0.1	Francesco	Introduction	Chapter 1, Sections 1-2; Chapter 2, Sections 1-2	A (rather dated) survey on the field of autonomous agents and multiagent systems is reported in: Special issue "Ten Years of Autonomous Agents and Multiagent Systems", AI Magazine, 33(3):11-105, Fall 2012. http://www.aaai.org/ojs/index.php/aimagazine/issue/view/199/showToc A recent paper that frames the new concepts "AI agents" and "agentic AI" in the context of previous results on agents and multi-agent systems is available here: https://arxiv.org/pdf/2506.01463 A blog post discussing some companies that apply multiagent approaches and technologies is available here: https://medium.com/@Francesco_AI/distributed-artificial-intelligence-3e3491e0771c A non-technical introduction to multiagent systems is available here (in Italian): https://www.agendadigitale.eu/cultura-digitale/i-sistemi-multiagente-cosi-governiamo-lintelligenza-artificiale/
Wednesday	16/9	14:15 - 16:15	3.1.8	Francesco	Agent architectures	Chapter 1, Section 3.2	
Tuesday	22/9	8:15 - 10:15	6.0.1	Francesco	Agent architectures	Chapter 1, Section 3.2	A description of Markov decision processes is reported in the Chapter 3 of the following book: Richard Sutton, Andrew Barto, Reinforcement Learning: An Introduction, second edition, MIT Press, Cambridge, MA, 2018. http://www.incompleteideas.net/book/the-book-2nd.html
Wednesday	23/9	14:15 - 16:15	3.1.8	Francesco	Exercises on agent architectures		
Tuesday	29/9	8:15 - 10:15	6.0.1		**NO CLASS**		
Wednesday	30/9	14:15 - 16:15	3.1.8	Francesco	Coalition formation	Chapter 8, Sections 1-3 (except Sections 3.2, 3.4-3.5)	
Tuesday	6/10	8:15 - 10:15	6.0.1	Nicola	Basics of game theory	Chapter 17	The general theme of the growing interactions between computer science and game theory is discussed in the following article: Yoav Shoham, "Computer Science and Game Theory", Communications of the ACM, 51 (8):74-79, August 2008. https://dl.acm.org/doi/10.1145/1378704.1378721 (The above document may be accessed, following the reported link, from within the Politecnico network.) A blog on Algorithmic Game Theory is: http://agtb.wordpress.com/ The hypothesis of rationality for human players is problematic. For an informal discussion, see for example: Kaushik Basu, "The Traveler's Dilemma", Scientific American, June 2007. http://www.cs.virginia.edu/~robins/The_Travelers_Dilemma.pdf An introduction to the iterated prisoner's dilemma is available at: http://en.wikipedia.org/wiki/Prisoner's_dilemma
Wednesday	7/10	14:15 - 16:15	3.1.8	Francesco	Coalition formation	Chapter 8, Section 5 (except Sections 5.3-5.6)	Notes on the distributed algorithm for finding the best coalition structure: https://amigoni.faculty.polimi.it/teaching/NotesAAMAS2017-2018.pdf
Tuesday	13/10	8:15 - 10:15	6.0.1	Nicola	Basics of game theory	Chapter 17	
Wednesday	14/10	14:15 - 16:15	3.1.8	Francesco	Exercises on game theory		
Tuesday	20/10	8:15 - 10:15	6.0.1	Francesco	Exercises on coalition formation		
Wednesday	21/10	14:15 - 16:15	3.1.8		**NO CLASS**		
Tuesday	27/10	8:15 - 10:15	6.0.1	Francesco	Multiagent planning	Chpater 11, Sections 1-2	Refer to Sections 2-4 of the following paper for an exhaustive algorithm that finds the optimal joint policy for a DEC-POMDP: R. Nair, M. Tambe, M. Yokoo, D. Pynadath, and S. Marsella, "Taming Decentralized POMDPs: Towards Efficient Policy Computation for Multiagent Settings", Proceedings of the International Joint Conference on Artificial Intelligence (IJCAI), p. 705-711, 2003. https://people.ict.usc.edu/~pynadath/Papers/ijcai03.pdf
Wednesday	28/10	14:15 - 16:15	3.1.8	Francesco	Multiagent planning	Chapter 11, Secrion 5 (except Section 5.3)	
Tuesday	3/11	8:15 - 10:15	6.0.1	Francesco	Multiagent planning		
Wednesday	4/11	14:15 - 16:15	3.1.8	Francesco	Exercises on multiagent planning		
Tuesday	10/11	8:15 - 10:15	6.0.1	Francesco	Distributed constraint optimization	Chapter 12, Sections 1-4 (except Section 4.1)	
Wednesday	11/11	14:15 - 16:15	3.1.8	Francesco	Distributed constraint optimization	Chapter 12, Section 5	
Tuesday	17/11	8:15 - 10:15	6.0.1	Francesco	Distributed constraint optimization		
Wednesday	18/11	14:15 - 16:15	3.1.8	Francesco	Exercises on distributed constrain optimization		
Tuesday	24/11	8:15 - 10:15	6.0.1	Francesco	Hands-on on distributed constrain optimization		
Wednesday	25/11	14:15 - 16:15	3.1.8	Nicola	Multiagent learning	Chapter 10, Sections 1-3	Refer to Sections 2-3, 5 of the following paper for a more complete illustration of stochastic games: Junling Hu and Michael Wellman, "Nash Q-Learning for General-Sum Stochastic Games", Journal of Machine Learning Research, 4:1039-1069, 2003. http://www.jmlr.org/papers/volume4/hu03a/hu03a.pdf
Tuesday	1/12	8:15 - 10:15	6.0.1	Nicola	Multiagent learning	Chapter 10, Section 4	
Wednesday	2/12	14:15 - 16:15	3.1.8	Nicola	Multiagent learning		
Tuesday	8/12	8:15 - 10:15	6.0.1		**NO CLASS**		
Wednesday	9/12	14:15 - 16:15	3.1.8	Nicola	Multiagent learning		
Tuesday	15/12	8:15 - 10:15	6.0.1	Francesco	Exercises on multiagent learning		
Wednesday	16/12	14:15 - 16:15	3.1.8		**NO CLASS**		
Tuesday	22/12	8:15 - 10:15	6.0.1	Francesco	Agentic Artificial Intelligence		
					Applications of multiagent systems		Slides on applications of agent systems in space by prof. Michèle Lavagna: https://amigoni.faculty.polimi.it/teaching/AgentsInSpace.pdf A NASA mission that is expected to bring the first multi-robot system to the Moon is described here: https://www.jpl.nasa.gov/missions/cadre/ An agent-based system for controlling a manufacturing plant is described in: Klaus Schild, Stefan Bussmann, "Self-Organization in Manufacturing Operations", Communications of the ACM, 50(12):74-79, December 2007. http://portal.acm.org/citation.cfm?id=1323688.1323698 (The above document may be accessed, following the reported link, from within the Politecnico network.) An interesting application of the agent paradigm (also in this case not implemented with agent technologies) is the High-Frequency-Trading that substitutes human brokers: http://arstechnica.com/tech-policy/news/2009/07/-it-sounds-like-something.ars